

Healthcare Analytics Ad-hoc Requests

Stakeholder Questions and SQL/Power BI Analysis Plan

Project Type	End-to-End Healthcare Analytics Portfolio Project
Dataset	Synthetic data on approximately 1,000 patients of Massachusetts General Hospital, 2011-2022
Tools	Excel, Python, SQL Server, Power BI
Document	Ad-hoc Requests PDF

1. Purpose of This Document

This document lists realistic ad-hoc stakeholder requests for the Massachusetts General Hospital synthetic patient-records project. These requests are designed to show how a data analyst translates business needs into measurable questions, SQL outputs, and Power BI visuals.

2. Dataset Context

The dataset contains synthetic records for approximately 1,000 patients from 2011-2022. It includes patient demographics, insurance and payer information, hospital encounters, procedures, claim costs, payer coverage, patient out-of-pocket cost, length of stay, and repeat-visit indicators.

3. Recommended Analysis Questions

- How many patients have been admitted or readmitted over time?
- How long are patients staying in the hospital, on average?
- How much is the average cost per visit?
- How many procedures are covered by insurance?

4. Top 10 Real-World Ad-hoc Requests

Request ID	Stakeholder	Ad-hoc Request	Expected Insight	BI Output
REQ-01	Hospital Management	How many total encounters occurred each year?	Yearly encounter trend showing hospital load from 2011-2022.	Line chart and KPI trend
REQ-02	Operations Team	For each year, what percentage of encounters belonged to each encounter class?	Encounter class mix by year for ambulatory, outpatient, urgent care, emergency, wellness, and inpatient.	Stacked bar or matrix
REQ-03	Operations Team	What percentage of encounters lasted over 24 hours versus 24 hours or less?	Short vs long-duration encounter split.	Donut chart and callout
REQ-04	Finance Team	How many encounters had zero payer coverage and what percentage does this represent?	Zero coverage count and percentage of all encounters.	KPI card and donut chart
REQ-05	Insurance Team	Which payer has the highest total coverage and which payer has the strongest coverage ratio?	Payer-wise total coverage and coverage ratio comparison.	Bar chart
REQ-06	Finance Team	What is the average total claim cost for encounters, broken down by payer?	Average claim cost by payer to identify costly payer categories.	Column chart
REQ-07	Clinical Operations	What are the top 10 most frequent procedures performed?	Procedure workload ranking by volume.	Treemap or bar chart
REQ-08	Finance + Clinical Leads	Which procedures generate the highest total cost?	Procedure-level financial impact, not just average cost.	Horizontal bar chart and matrix
REQ-09	Care Coordination Team	How many patients had a 30-day repeat encounter?	Unique patient count and total repeat encounters within 30 days.	KPI and donut chart

Request ID	Stakeholder	Ad-hoc Request	Expected Insight	BI Output
REQ-10	Care Coordination Team	Which patients had the most repeat visits?	High-utilization patient list for care coordination planning.	Table sorted by repeat visit count

5. Mapping Ad-hoc Requests to Dashboard Pages

Dashboard Page	Requests Covered
Home Page	Project summary, dataset metrics, navigation, and workflow.
Operations Overview	REQ-01, REQ-02, REQ-03: hospital load, encounter mix, monthly trend, stay duration.
Cost & Coverage	REQ-04, REQ-05, REQ-06: total claim cost, payer coverage, patient responsibility, zero-coverage risk.
Procedure Analysis	REQ-07, REQ-08: procedure volume, average cost, total cost, top procedure ranking.
Readmission Analysis	REQ-09, REQ-10: 30-day repeat visits, demographics, high-utilization patients.

6. Priority Ranking

Priority	Business Focus
High	Zero payer coverage, total patient responsibility, high-utilization patients, 30-day repeat encounter rate
Medium	Procedure workload, top procedure cost, average claim cost by payer, encounter class mix
Supporting	Monthly pattern, quarterly patient trend, stay duration split

7. How This Helps in Interviews

This ad-hoc request document shows that the project is not only a technical dashboard but a stakeholder-driven analytics case study. Each question connects to a business user, a measurable output, and a dashboard visual. This makes the project easier to explain in interviews and stronger for GitHub portfolio presentation.